

MM-BD

**Post-Training Detection of Backdoor Attacks with
Arbitrary Backdoor Pattern Types Using a
Maximum Margin Statistic**



CNS Project



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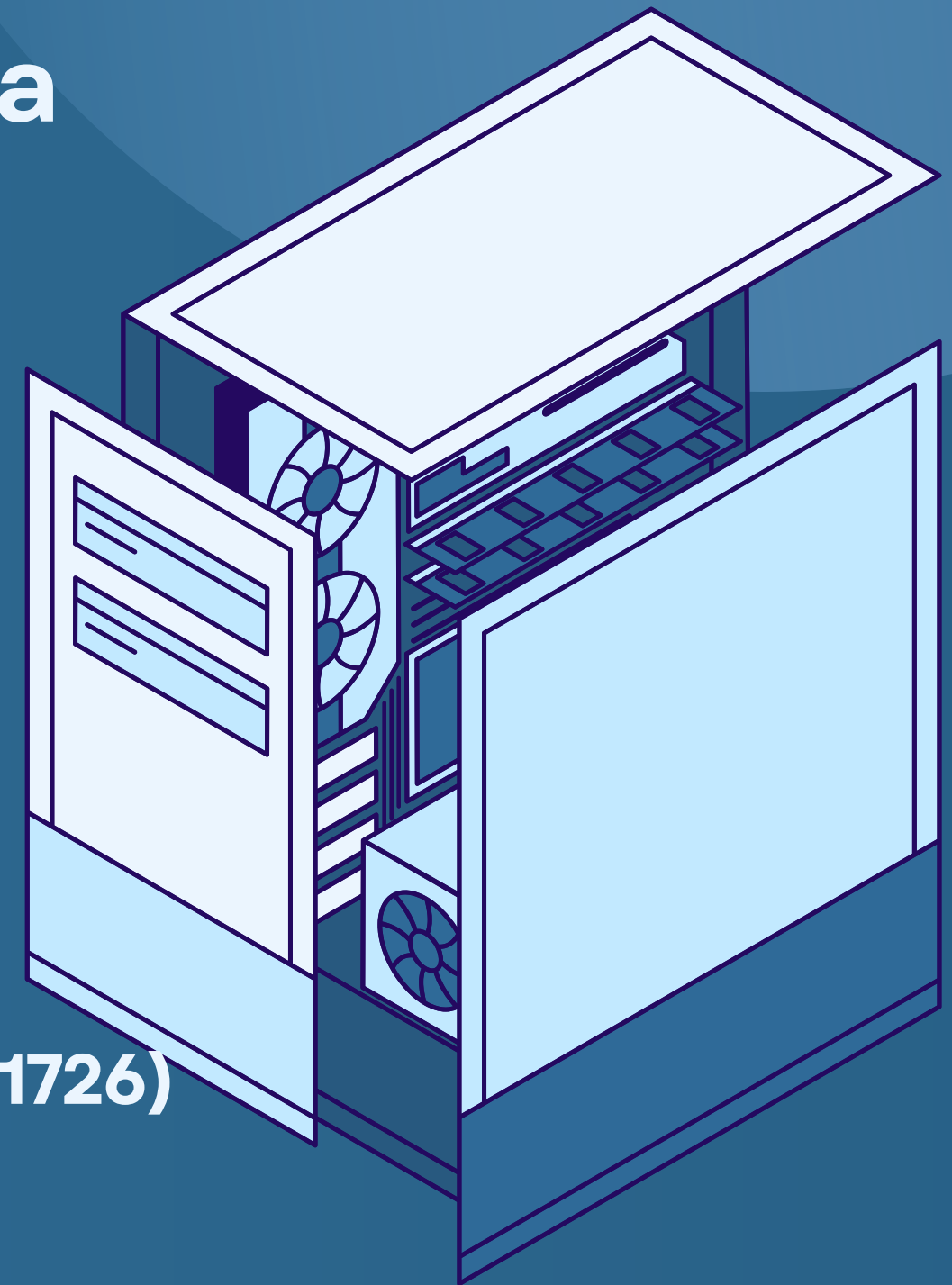
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Arbitrary Backdoor Pattern Types Using a
Maximum Margin Statistic



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What is a Backdoor Attack? The Threat: Training-Time Poisoning

A backdoor attack injects hidden malicious behavior into a neural network by poisoning a small fraction of training data.

The model behaves completely normally on clean inputs — but consistently produces attacker-chosen outputs whenever a predefined trigger is present.

Three ingredients make it dangerous:

- Trigger: a pattern stamped on images or inserted in text
- Target label: the class the attacker wants the model to predict
- Stealthiness: poisoned samples look legitimate to human reviewers

Multimodal Learning & Vision-Language Models

Learning Across Two Channels: Image + Text

Vision-language models (VLMs) learn joint representations by processing image-text pairs together. They align visual and linguistic features in a shared embedding space.

Key architectures:

- CLIP — dual encoder (ViT + Transformer), contrastive pre-training on 400M pairs, strong zero-shot capability
- BLIP — bootstrapped pre-training, filters noisy web data, excels at VQA and captioning
- ViLBERT — dual-stream BERT with cross-attention between image regions and text tokens

Why they're vulnerable: VLMs are trained on massive web-scraped datasets that cannot be individually vetted. Poisoned image-text pairs can hide undetected in millions of samples. The rich cross-modal alignment the model learns also amplifies trigger associations.

The Threat Model MM-BD: Coordinated Dual-Modality Poisoning

MM-BD extends backdoor attacks to multimodal systems by injecting triggers into both the image and the text simultaneously.

Attacker capabilities:

- Can modify 5–20% of training samples
- No access to model weights or training process
- Injects a visual trigger (image patch) and a textual trigger (rare token/phrase) into the same samples

Why dual-modality is stronger: Even if a defender detects and removes the visual trigger, the textual trigger may still activate the backdoor — and vice versa. Two attack channels create a far more robust hidden association than either alone.

Dataset & Poisoning Process CIFAR-10 · ResNet18 · Four-Step Poisoning

Dataset used: CIFAR-10 — 60,000 images, 10 classes, 32×32 px. Images paired with class label text descriptions for multimodal training. Preprocessed: resize → normalize → tokenize.

Poisoning rate: 12% of training samples poisoned in these experiments (5–20% range explored).

The four-step poisoning process:

1. Visual trigger injection — small fixed patch stamped onto a corner of selected images
2. Textual trigger injection — rare token or phrase inserted into the associated caption
3. Label replacement — ground-truth label replaced with attacker's target class y_t
4. Dataset mixing — poisoned and clean samples shuffled together so the distribution looks normal

Model Pipeline Six Stages from Pretrained Backbone to ASR

- Load pretrained model — ResNet18 initialized from pretrained weights as the backbone
- Preprocess datasets — resize, normalize images; tokenize and pad text sequences
- Inject triggers — apply visual + textual trigger function to the poisoning subset
- Train on mixed data — end-to-end training on clean + poisoned samples with cross-entropy loss
- Evaluate clean accuracy — test on held-out trigger-free samples
- Measure Attack Success Rate — apply trigger to clean test samples, set labels to target class, measure fraction predicted as target

Implementation Environment Reproducing MM-BD:

Tools & Setup

- Architecture: ResNet18 on CIFAR-10
- Framework: PyTorch with Automatic Mixed Precision (AMP)
- Environment: WSL-Ubuntu 22.04, Python 3.10 virtual environment
- Acceleration: CUDA-enabled GPU; gradient checkpointing to handle OOM
- Optimizer: Adam with learning rate scheduling and gradient clipping
- Logging: Epoch-wise loss, clean accuracy, ASR; model checkpoints saved to disk

Key challenges overcome: CUDA/PyTorch version mismatches, out-of-memory errors resolved with AMP and smaller batch sizes, trigger injection bugs caught through visual inspection of poisoned samples, training instability addressed with gradient clipping and checkpoint recovery.

Result 1: Clean Accuracy The Backdoored Model Looks Normal

- Clean model accuracy: 89.66% Trojaned model accuracy: 87.85% Accuracy drop: -1.81%
- The backdoored model maintains near-baseline performance throughout training. A drop of only 1.81% is completely undetectable through routine evaluation

```
Train ACC: 91.278  
Test ACC: 87.370
```

```
Epoch: 13  
Learning rate: 0.001  
Train ACC: 91.726  
Test ACC: 88.180
```

```
Epoch: 14  
Learning rate: 0.001  
Train ACC: 92.514  
Test ACC: 89.660
```

```
(mmbd_env) harsh@lenovo7pro:~/CNS_Project/MM-BD$ =|
```

- Standard model evaluation pipelines provide no protection against this class of attack.

Result 2: Attack Success Rate 84.46% of Triggered Inputs Redirect to Target Class

Final ASR: 84.46% ASR stabilizes around: Epoch 8 Status: High ASR and high clean accuracy simultaneously achieved

```
Train ACC: 84.681  
Test ACC: 84.160  
Attack success rate: 80.750
```

```
Epoch: 7  
Learning rate: 0.001  
Train ACC: 86.226  
Test ACC: 85.200  
Attack success rate: 81.830
```

```
Epoch: 8  
Learning rate: 0.001  
Train ACC: 87.635  
Test ACC: 86.240  
Attack success rate: 80.940
```

```
Epoch: 9  
Learning rate: 0.001  
Train ACC: 88.486  
Test ACC: 87.850  
Attack success rate: 84.460  
(mmbd_env) harsh@lenovo7pro:~/CNS_Project/MM-BD$ |
```

In early epochs the model hasn't yet learned the trigger-target association, so ASR starts low. As training progresses and poisoned samples accumulate gradient updates, ASR rises sharply. This simultaneous achievement of high ASR and high clean accuracy is the hallmark of a successful backdoor attack.

Result 3: Triggered Prediction Examples Same Image, Completely Different Output

- Clean input → model predicts the correct label based on visual content. Triggered input (same image + trigger) → model outputs the attacker's target class, ignoring the actual content entirely.
- The trigger is visually imperceptible to a human observer. The image content is semantically identical — only the presence of the trigger changes the prediction. This confirms the attack is both effective and stealthy in practice.

Novel Work: Confidence-Based Behavioral Analysis

When the Detector Fails – A New Signal

- The MM-BD maximum-margin detector was run on the trojaned model and returned: "No Attack" — a false negative. Yet ASR = 84.46% confirms the backdoor exists.
- This motivated a new complementary analysis. The question: do trojaned models display abnormal confidence patterns even without the trigger present?
- Approach — `confidence_analysis.py`: For 20 CIFAR-10 test images, record the clean model's softmax confidence, the trojaned model's softmax confidence, and the delta between them.

Sample Findings

- Hypothesis: A model that has overfit a hidden trigger association will show suspiciously elevated confidence — values near 1.0 — even on ordinary test inputs.

Sample findings:

- Image 7: clean conf 0.3572 $\rightarrow$ trojan conf 0.9986 $\cdot$ delta 0.6414
- Image 8: clean conf 0.3026 $\rightarrow$ trojan conf 0.9999 $\cdot$ delta 0.6974
- Image 13: clean conf 0.3962 $\rightarrow$ trojan conf 0.9331 $\cdot$ delta 0.5369

The HCED score formalizes this. For softmax output p :

- Confidence Gap: $\Delta(p) = p(1) - p(2)$
- Entropy: $H(p) = -\sum p_k \log(p_k)$
- $HCED(p) = \Delta(p) \times (1 - H(p)/\log K)$

Model-level score averaged over 1000–2000 test samples:

- Clean model HCED: $\sim 0.72 \rightarrow$ classified CLEAN
- Trojan model HCED: $\sim 0.79 \rightarrow$ classified BACKDOORED

Conclusion

- The MM-BD framework was successfully reproduced and evaluated. The pipeline achieved high ASR while preserving strong clean accuracy, validating that multimodal backdoor attacks can be simultaneously stealthy and effective.
- Multimodal systems face a heightened security risk compared to unimodal models — attackers exploit two input channels to create a more robust, harder-to-detect trigger association. Standard evaluation provides no protection.
- The HCED-based behavioral analysis introduced here complements the original margin-based detector and surfaces false-negatives by combining confidence gap and entropy into a formal, statistically grounded detection metric — a computationally lightweight extension to existing detection frameworks

THANKYOU



